Veronica Malewski
CS 340
Project Two
October 25, 2026

Southern New Hampshire University

CS 340 README

About the Project

The AAC Animal Shelter project uses Python, MongoDB, and Dash to create a dashboard for Grazioso Salvare. The dashboard connects to a MongoDB database containing Austin Animal Center Outcomes data (Austin Animal Center, 2020) using the CRUD Python module created in Project One. The dashboard allows the user to view animal records and filter the data for Water Rescue, Mountain or Wilderness Rescue, Disaster Rescue or Individual Tracking, or Reset. The data table, pie chart, and map update based on the selected filter.

Motivation

The goal of this project was to create a dashboard that makes the animal shelter data easier for Grazioso Salvare to use. Instead of searching through the MongoDB database manually, the dashboard gives the user simple filter options and displays the results through an interactive data table, pie chart, and geolocation map. The CRUD module also keeps the database connection and queries reusable instead of repeating the same MongoDB code throughout the dashboard. Each rescue filter updates the dashboard components so Grazioso Salvare can quickly identify animals that match specific rescue-training requirements. This supports the project goal of creating a user-friendly dashboard that makes the data easier to access, understand, and maintain.

Getting Started

This project was completed in Codio using Jupyter Lab. To set up the project, the Austin Animal Center Outcomes data should be imported into an `aac` database with an animals

collection, and the aacuser account should have access to the database. Add CRUD_Python_Module.py and ProjectTwoDashboard.ipynb to the project files and make sure the Grazioso Salvare logo is available in the code_files directory.

The CRUD module connects the dashboard to MongoDB using PyMongo, while the notebook uses Dash to display and filter the animal data. Open ProjectTwoDashboard.ipynb in Jupyter Lab and run the notebook to start the dashboard. The radio buttons can then be used to switch between the rescue filters.

Installation

This project was completed in Codio, where most of the required tools were already available. Python was used for the dashboard and CRUD module, MongoDB stored the animal records, and PyMongo connected Python to MongoDB. Jupyter Lab was used to run the notebook, while Dash, Plotly, and Dash Leaflet were used to create the dashboard components. A local setup would require the main tools: Python, MongoDB, PyMongo, Dash, Plotly, Jupyter Lab, and Dash Leaflet.

Resources Used:

- Python: https://www.python.org/
- MongoDB: https://www.mongodb.com/
- Dash: https://dash.plotly.com/
- Plotly: https://plotly.com/python/
- Jupyter: https://jupyter.org/

Tools Used

1. **MongoDB**

MongoDB was used as the model portion of the project because it stores the Austin Animal Center records and supports queries based on fields such as breed, sex, and age. It also works with Python through PyMongo, which allowed the dashboard to retrieve records using the CRUD module.

2. **PyMongo and the CRUD Module**

PyMongo was used to connect the Python code to MongoDB. The AnimalShelter class from Project One contains the CRUD methods used to work with the database. For the dashboard, the Read method retrieves either the full data set or records that match one of the rescue filters. MongoDB's PyMongo documentation explains that PyMongo allows Python applications to connect to MongoDB and retrieve matching records using methods such as find() (MongoDB, Inc., n.d.).

3. **Dash**

Dash was used to create the view and controller portions of the application. The dashboard includes radio buttons, a data table, a pie chart, and a map. Dash callbacks connect these components so that selecting a filter updates the table and charts.

4. **Plotly and Dash Leaflet**

Plotly Express was used to create the pie chart showing the most common breeds in the current results. Dash Leaflet was used to create the geolocation map using the latitude and longitude values in the animal records.

Usage

When the dashboard first opens, it displays the unfiltered Austin Animal Center data. The data table shows 10 records per page and includes sorting and pagination. Each rescue option sends a MongoDB query based on the breed, sex, and age requirements provided by Grazioso Salvare. The table, pie chart, and map update when a different filter is selected. Reset returns the dashboard to the full unfiltered data set. The map also responds to the selected row in the table and displays the location of that animal. These features follow the functionality outlined in the Dashboard Specifications Document (Southern New Hampshire University, n.d.-a).

The user can select one of four options:

- Water Rescue
- Mountain or Wilderness Rescue
- Disaster Rescue or Individual Tracking
- Reset

Steps Taken

I first reused the CRUD Python module from Project One to connect the dashboard to MongoDB. I used the Read method with an empty query to retrieve all animal records for the starting dashboard. Next, I added the required radio buttons and created MongoDB queries for each rescue category using the breed, sex, and age requirements from the Dashboard Specifications Document (Southern New Hampshire University, n.d.-a). I connected the filters to the data table with a Dash callback so the table would update when a different option was selected. I then added a pie chart and connected the geolocation map from the Module Six milestone (Southern New Hampshire University, n.d.-c). Finally, I added the Grazioso Salvare

logo, my name, pagination, sorting, and the final dashboard layout. I tested the starting state and each filter to make sure the data table, pie chart, and map updated correctly.

Challenges

One challenge was matching the breed requirements to the actual values stored in the database. Some animals were listed as *German Shepherd Mix* instead of only *German Shepherd*. I used regular expression searches in the MongoDB queries so those breed variations would still be included.

Another issue was getting the Grazioso Salvare logo to load because the notebook and the logo were stored in different folders. I corrected the relative file path so the notebook could find and display the image.

The unfiltered pie chart was also difficult to read because there were many different breeds in the full data set. I changed the chart to display the ten most common breeds and adjusted the bottom layout to give more space to the map.

Tests

I tested the dashboard throughout development instead of waiting until the end. As each feature was added, I ran the notebook and checked that it worked before moving on to the next part. This helped catch problems early, such as the logo file path, filter results, and chart layout. Microsoft recommends testing early in the software development lifecycle and making testing an ongoing part of the development process, which is the approach I followed while building the dashboard (Microsoft, n.d.). I also completed final testing of the starting state and each required filter as required for Project Two (Southern New Hampshire University, n.d.-b).

1. Starter State



CS-340 Dashboard

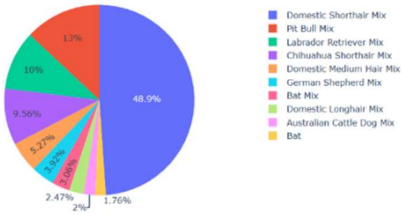
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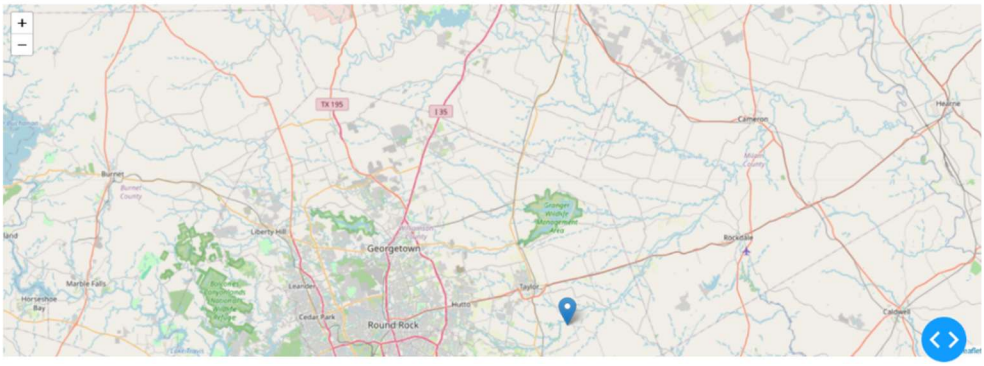
○ Water Rescue ○ Mountain or Wilderness Rescue ○ Disaster Rescue or Individual Tracking ◉ Reset

	rec_num	age_upon_outcome	animal_id	animal_type	breed	color	date_of_birth	datetime	monthyear	name	outcome_subtype	outcome_type	sex_upon_outcome	location_lat	location_long	age_upon_outcome_in_weeks
◉	1	3 years	A746874	Cat	Domestic Shorthair Mix	Black/White	2014-04-10	2017-04-11 09:00:00	2017-04-11T09:00:00		SCRP	Transfer	Neutered Male	30.5066578739455	-97.3408780722188	156.767857142857
○	9	3 years	A720214	Dog	Labrador Retriever Mix	Red/White	2013-02-04	2016-02-11 12:41:00	2016-02-11T12:41:00	Blessing		Adoption	Spayed Female	30.3870648199411	-97.3684339731375	157.504067460317
○	3	2 years	A716330	Dog	Chihuahua Shorthair Mix	Brown/White	2013-11-18	2015-12-28 18:43:00	2015-12-28T18:43:00	Frank		Adoption	Neutered Male	30.7595748121648	-97.5523753807133	110.111408730159
○	10	3 months	A664290	Cat	Domestic Shorthair Mix	Tortie	2013-09-01	2013-12-08 14:58:00	2013-12-08T14:58:00	*Taylor		Adoption	Spayed Female	30.7583105481048	-97.618292198845	14.0890873015873
○	11	1 year	A721199	Dog	Dachshund Wirehair Mix	Tan/White	2015-02-23	2016-02-27 17:49:00	2016-02-27T17:49:00	Belle		Adoption	Spayed Female	30.7290272761146	-97.3753328216134	52.820337301587
○	12	1 year	A664843	Dog	Pit Bull Mix	Brown/White	2013-06-09	2014-08-18 17:24:00	2014-08-18T17:24:00	Sherlock	Partner	Transfer	Neutered Male	30.4515549397366	-97.474104510925	62.2464285714288
○	13	1 year	A700408	Cat	Domestic Shorthair Mix	Brown Tabby/White	2014-04-13	2015-04-15 13:34:00	2015-04-15T13:34:00	Nyla		Return to Owner	Spayed Female	30.4101154527976	-97.562415670838	52.5093253968254
○	14	2 years	A742287	Dog	Boxer/Bullmastiff	Brown Brindle/White	2015-01-18	2017-02-11 12:30:00	2017-02-11T12:30:00	*Kawhi		Adoption	Neutered Male	30.4551148649096	-97.3087780473978	107.931547619048
○	5	2 years	A691584	Dog	Labrador Retriever Mix	Tan/White	2012-11-06	2015-05-30 13:48:00	2015-05-30T13:48:00	Luke		Return to Owner	Neutered Male	30.7104815618433	-97.562297435286	133.653571428571
○	15	3 years	A712638	Dog	Pit Bull Mix	Red/White	2012-09-26	2016-07-18 17:52:00	2016-07-18T17:52:00	Marcus	Partner	Transfer	Neutered Male	30.5798299207017	-97.5588487936533	198.820634920635

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Preferred Animals





2. Water Rescue



CS-340 Dashboard

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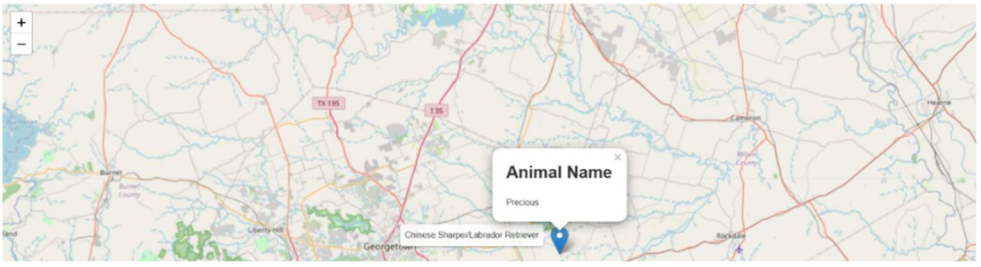
◉ Water Rescue ○ Mountain or Wilderness Rescue ○ Disaster Rescue or Individual Tracking ○ Reset

	rec_num	age_upon_outcome	animal_id	animal_type	breed	color	date_of_birth	datetime	monthyear	name	outcome_subtype	outcome_type	sex_upon_outcome	location_lat	location_long	age_upon_outcome_in_weeks
◉	3905	1 year	A750806	Dog	Chinese Sharpei/Labrador Retriever	Chocolate	2016-05-31	2017-09-26 14:46:00	2017-09-26T14:46:00	Precious		Return to Owner	Intact Female	30.6229710455683	-97.3445078033179	69.0878968253968
○	6128	9 months	A745478	Dog	German Shepherd/Labrador Retriever	Tricolor	2016-06-18	2017-03-24 12:03:00	2017-03-24T12:03:00	*Myra	Partner	Transfer	Intact Female	30.7007209771159	-97.247241975955	39.928869047619
○	4928	2 years	A764876	Dog	Labrador Retriever	Yellow	2016-01-08	2018-01-08 12:52:00	2018-01-08T12:52:00	Rosa		Return to Owner	Intact Female	30.4973908876274	-97.287076323936	104.505158730159
○	36	6 months	A706953	Dog	Labrador Retriever Mix	Yellow	2014-12-06	2015-07-06 11:33:00	2015-07-06T11:33:00		Medical	Euthanasia	Intact Female	30.5480802368633	-97.2969969058957	30.3544642857143
○	732	2 years	A749782	Dog	Labrador Retriever Mix	Tan/White	2015-05-19	2017-07-25 14:59:00	2017-07-25T14:59:00	*Catalina		Return to Owner	Intact Female	30.6138310636757	-97.5752164857665	114.089186507937
○	1121	1 year	A757158	Dog	Labrador Retriever Mix	White/Black	2016-08-30	2017-08-31 14:12:00	2017-08-31T14:12:00	Pirata		Return to Owner	Intact Female	30.5572161697962	-97.5363224263878	52.3702380952381
○	1628	9 months	A740471	Dog	Labrador Retriever Mix	Tan/White	2016-03-17	2016-12-23 17:13:00	2016-12-23T17:13:00	Mika		Adoption	Intact Female	30.7569243032341	-97.7392549176654	40.2453373015873
○	1757	7 months	A742767	Dog	Labrador Retriever Mix	Black	2016-06-27	2017-02-14 15:20:00	2017-02-14T15:20:00	Marley		Return to Owner	Intact Female	30.4869754937324	-97.4280017197358	33.234126984127
○	1988	1 year	A762781	Dog	Labrador Retriever Mix	Black/White	2016-11-27	2017-12-03 13:09:00	2017-12-03T13:09:00		Partner	Transfer	Intact Female	30.2840111162863	-97.4600542219677	53.0782738095238
○	2041	2 years	A702745	Dog	Labrador Retriever Mix	Black	2013-05-22	2015-05-22 11:45:00	2015-05-22T11:45:00	Abigail		Return to Owner	Intact Female	30.7157941956301	-97.4523664870572	104.355654761905

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Preferred Animals





3. Mountain or Wilderness Rescue

CS-340 Dashboard

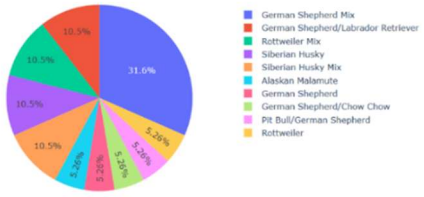
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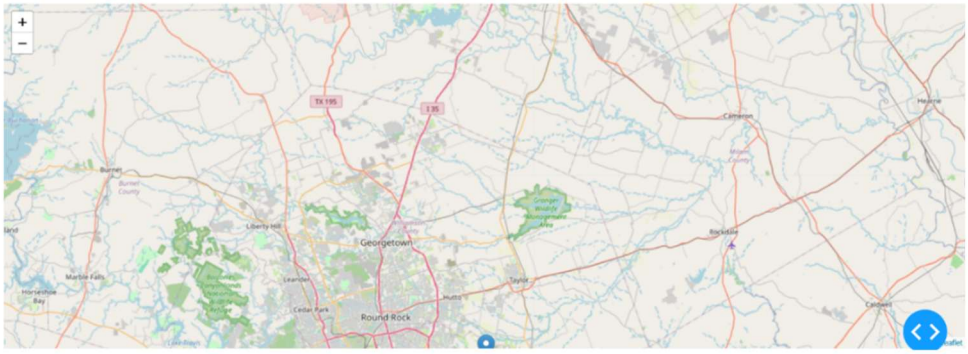
○ Water Rescue ◉ Mountain or Wilderness Rescue ○ Disaster Rescue or Individual Tracking ○ Reset

	rec_num	age_upon_outcome	animal_id	animal_type	breed	color	date_of_birth	datetime	monthyear	name	outcome_subtype	outcome_type	sex_upon_outcome	location_lat	location_long	age_upon_outcome_in_weeks
◉	5315	2 years	A708726	Dog	Alaskan Malamute	Sable/White	2013-07-30	2015-08-02 17:24:00	2015-08-02T17:24:00	Papa		Return to Owner	Intact Male	30.4309339291938	-97.480825836737	104.817857142857
○	6557	6 months	A765461	Dog	German Shepherd	Sable	2017-07-20	2018-01-22 11:54:00	2018-01-22T11:54:00	Sargent		Return to Owner	Intact Male	30.40668985085	-97.485680334264	26.6422619047619
○	2275	1 year	A668767	Dog	German Shepherd Mix	Black/Tan	2012-11-09	2013-12-09 18:29:00	2013-12-09T18:29:00	Vincent		Return to Owner	Intact Male	30.4693842352403	-97.2765316853847	56.5385912698413
○	5032	2 years	A765283	Dog	German Shepherd Mix	Black/Tan	2015-01-16	2017-11-05 17:05:00	2017-11-05T17:05:00	Lu		Adoption	Intact Male	30.5180710510914	-97.5035088748138	146.387400793651
○	5273	5 months	A730258	Dog	German Shepherd Mix	Black/Brown	2016-01-01	2016-07-01 13:36:00	2016-07-01T13:36:00	Teddy		Return to Owner	Intact Male	30.3238472197847	-97.3589995986214	26.0809523809524
○	5433	2 years	A673611	Dog	German Shepherd Mix	Black/Brown	2012-02-27	2014-03-07 15:46:00	2014-03-07T15:46:00	Blackie	Aggressive	Euthanasia	Intact Male	30.6984312942892	-97.6617369292416	105.665277777778
○	9544	1 year	A694389	Dog	German Shepherd Mix	Sable	2013-06-27	2014-12-27 16:19:00	2014-12-27T16:19:00	Coby		Return to Owner	Intact Male	30.2696033963098	-97.7026856199769	78.3828373015873
○	9655	1 year	A674720	Dog	German Shepherd Mix	Black	2012-09-23	2014-03-23 17:44:00	2014-03-23T17:44:00	Pachuco		Return to Owner	Intact Male	30.3695644651861	-97.6787136367287	78.1055555555556
○	5607	2 years	A669842	Dog	German Shepherd/Chow Chow	Black/Brown	2011-12-28	2014-01-03 14:25:00	2014-01-03T14:25:00	*Teddy	Partner	Transfer	Intact Male	30.5912312947521	-97.7343949868642	105.371527777778
○	1233	7 months	A722362	Dog	German Shepherd/Labrador Retriever	Tan/White	2015-08-25	2016-03-25 12:11:00	2016-03-25T12:11:00	Brisket		Return to Owner	Intact Male	30.5288318820223	-97.6407647288868	30.5010912698413

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Preferred Animals





4. Disaster Rescue or Individual Tracking

CS-340 Dashboard

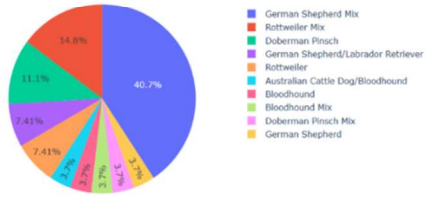
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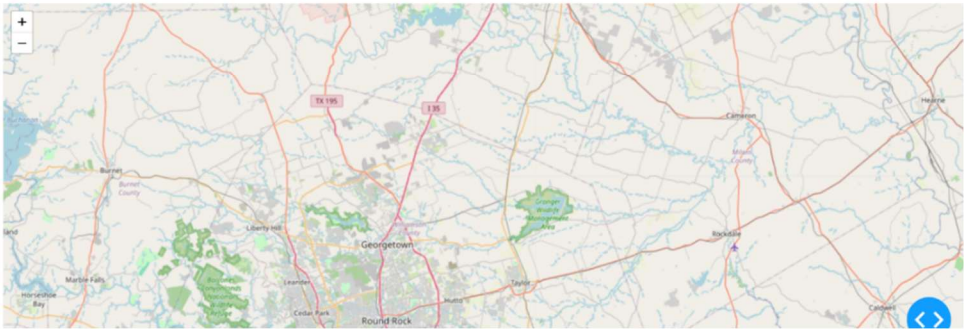
○ Water Rescue ○ Mountain or Wilderness Rescue ◉ Disaster Rescue or Individual Tracking ○ Reset

	rec_num	age_upon_outcome	animal_id	animal_type	breed	color	date_of_birth	datetime	monthyear	name	outcome_subtype	outcome_type	sex_upon_outcome	location_lat	location_long	age_upon_outcome_in_wee
◉	5411	5 months	A669643	Dog	Australian Cattle Dog/Bloodhound	Brown Brindle/White	2013-07-24	2014-01-05 13:10:00	2014-01-05T13:10:00	Dale		Return to Owner	Intact Male	30.2852781393318	-97.3924381749523	23.64980158730
○	3767	4 years	A712291	Dog	Bloodhound	Red	2011-09-20	2015-09-22 15:43:00	2015-09-22T15:43:00	Boomer		Return to Owner	Intact Male	30.2709983761287	-97.5923916912722	209.0935515873
○	1918	3 years	A698931	Dog	Bloodhound Mix	Red/Black	2012-03-19	2015-03-20 11:24:00	2015-03-20T11:24:00			Return to Owner	Intact Male	30.3638313841587	-97.3930177615336	156.6392857142
○	622	2 years	A741441	Dog	Doberman Pinsch	Blue/Tan	2015-01-04	2017-01-04 14:02:00	2017-01-04T14:02:00	Deeohgee		Return to Owner	Intact Male	30.6176290949888	-97.4249284729106	104.5121031746
○	4723	8 months	A731795	Dog	Doberman Pinsch	Black/Tan	2015-11-15	2016-08-06 21:42:00	2016-08-06T21:42:00	Negro		Return to Owner	Intact Male	30.5663297388775	-97.6449426083984	37.98630952380
○	7840	1 year	A654132	Dog	Doberman Pinsch	Black/Red	2012-07-16	2014-04-18 11:11:00	2014-04-18T11:11:00	Zorro	Partner	Transfer	Intact Male	30.760756173511	-97.3695237014692	91.6379960317
○	3964	2 years	A751717	Dog	Doberman Pinsch Mix	Black/Brown	2015-06-11	2017-06-12 12:32:00	2017-06-12T12:32:00	Rosco		Return to Owner	Intact Male	30.7107589510265	-97.5149092405164	104.6460317460
○	6557	6 months	A765461	Dog	German Shepherd	Sable	2017-07-20	2018-01-22 11:54:00	2018-01-22T11:54:00	Sargent		Return to Owner	Intact Male	30.40668985085	-97.485680334264	26.64226190476
○	256	3 years	A717863	Dog	German Shepherd Mix	Brown	2012-12-16	2015-12-20 00:00:00	2015-12-20T00:00:00	*Lucky	Partner	Transfer	Intact Male	30.5982744070273	-97.7012578871518	1
○	1765	3 years	A757370	Dog	German Shepherd Mix	Tricolor	2014-09-01	2017-10-03 13:28:00	2017-10-03T13:28:00	42	Partner	Transfer	Intact Male	30.2900119597318	-97.4421053357595	161.2230158730

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Preferred Animals





5. Reset

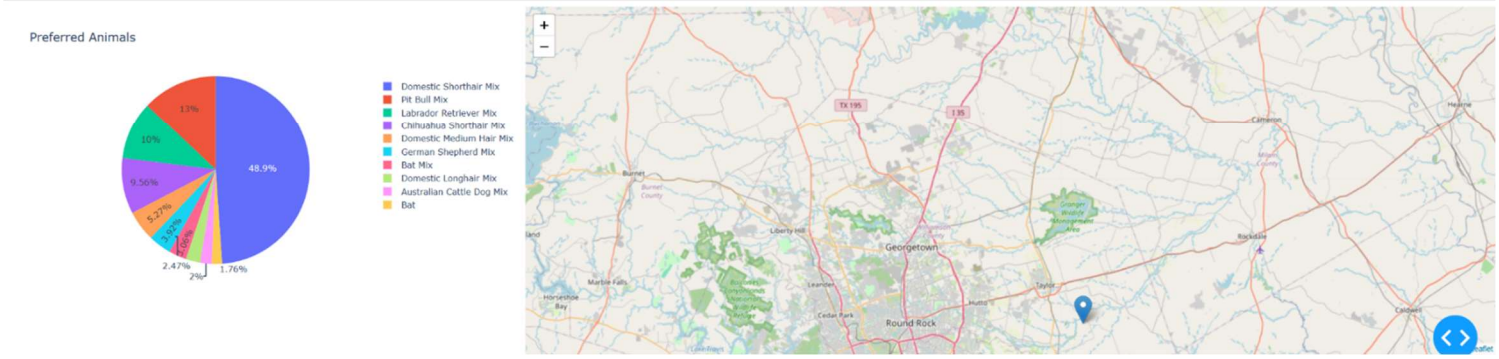
CS-340 Dashboard

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○ Water Rescue ○ Mountain or Wilderness Rescue ○ Disaster Rescue or Individual Tracking ◉ Reset

	rec_num	age_upon_outcome	animal_id	animal_type	breed	color	date_of_birth	datetime	monthyear	name	outcome_subtype	outcome_type	sex_upon_outcome	location_lat	location_long	age_upon_outcome_in_weeks
◉	1	3 years	A746874	Cat	Domestic Shorthair Mix	Black/White	2014-04-10	2017-04-11 09:00:00	2017-04-11T09:00:00		SCRP	Transfer	Neutered Male	30.5066578739455	-97.3408780722188	156.767857142857
○	9	3 years	A720214	Dog	Labrador Retriever Mix	Red/White	2013-02-04	2016-02-11 12:41:00	2016-02-11T12:41:00	Blessing		Adoption	Spayed Female	30.3870648199411	-97.3684339731375	157.504067460317
○	3	2 years	A716330	Dog	Chihuahua Shorthair Mix	Brown/White	2013-11-18	2015-12-28 18:43:00	2015-12-28T18:43:00	Frank		Adoption	Neutered Male	30.7595748121648	-97.5523753807133	110.111408730159
○	10	3 months	A664290	Cat	Domestic Shorthair Mix	Tortie	2013-09-01	2013-12-08 14:58:00	2013-12-08T14:58:00	*Taylor		Adoption	Spayed Female	30.7583105481048	-97.618292198845	14.0890873015873
○	11	1 year	A721199	Dog	Dachshund Wirehair Mix	Tan/White	2015-02-23	2016-02-27 17:49:00	2016-02-27T17:49:00	Belle		Adoption	Spayed Female	30.7290272761146	-97.3753328216134	52.8203373015873
○	12	1 year	A664843	Dog	Pit Bull Mix	Brown/White	2013-06-09	2014-08-18 17:24:00	2014-08-18T17:24:00	Sherlock	Partner	Transfer	Neutered Male	30.4515549397366	-97.474104510925	62.2464285714286
○	13	1 year	A700408	Cat	Domestic Shorthair Mix	Brown Tabby/White	2014-04-13	2015-04-15 13:34:00	2015-04-15T13:34:00	Nyla		Return to Owner	Spayed Female	30.4101154527976	-97.562415670838	52.5093253968254
○	14	2 years	A742287	Dog	Boxer/Bullmastiff	Brown Brindle/White	2015-01-18	2017-02-11 12:30:00	2017-02-11T12:30:00	*Kawhi		Adoption	Neutered Male	30.4551148649096	-97.3087780473978	107.931547619048
○	5	2 years	A691584	Dog	Labrador Retriever Mix	Tan/White	2012-11-06	2015-05-30 13:48:00	2015-05-30T13:48:00	Luke		Return to Owner	Neutered Male	30.7104815618433	-97.56229743528	133.653571428571
○	15	3 years	A712638	Dog	Pit Bull Mix	Red/White	2012-09-26	2016-07-18 17:52:00	2016-07-18T17:52:00	Marcus	Partner	Transfer	Neutered Male	30.5798299207017	-97.5588487936533	198.820634920635

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